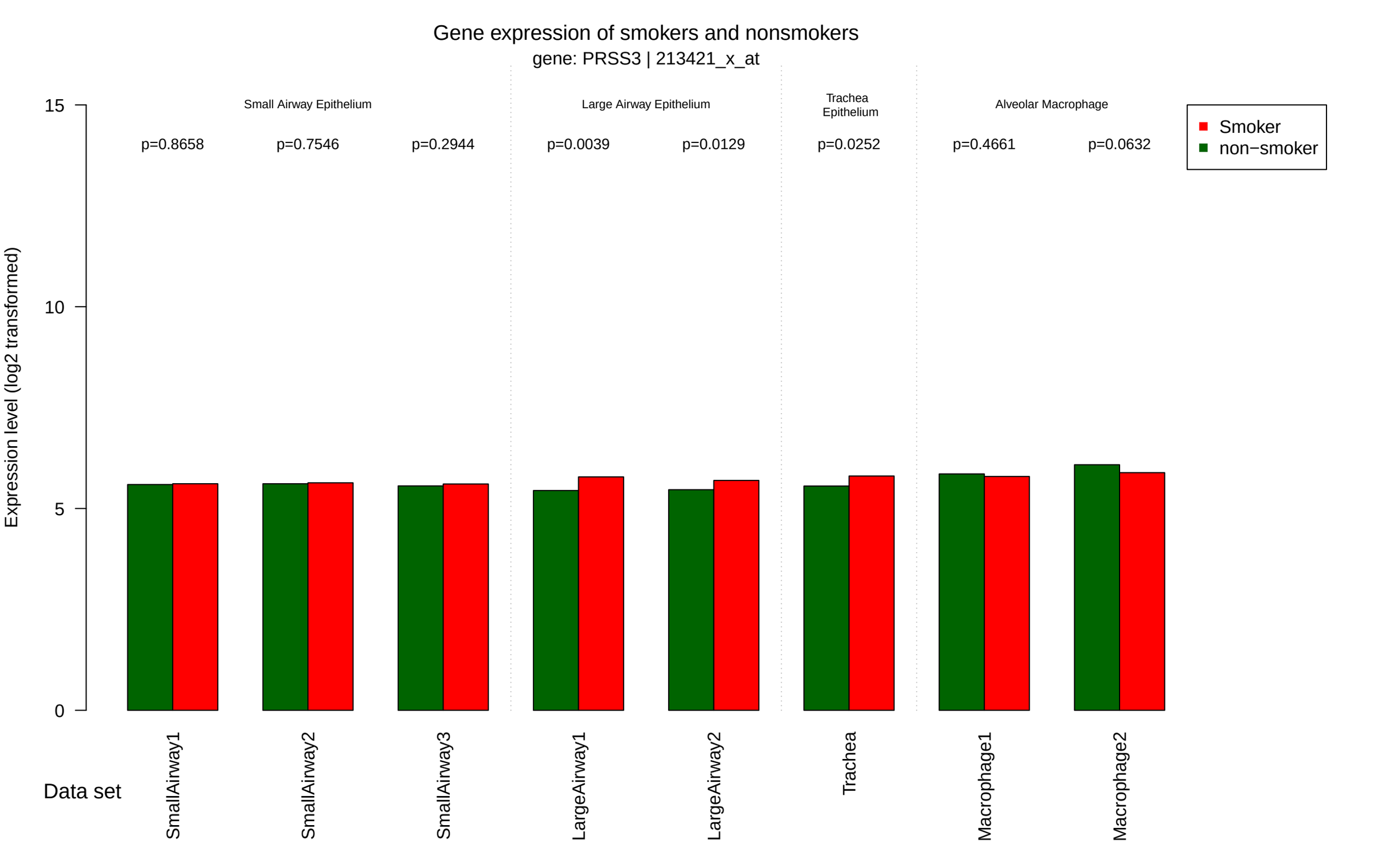
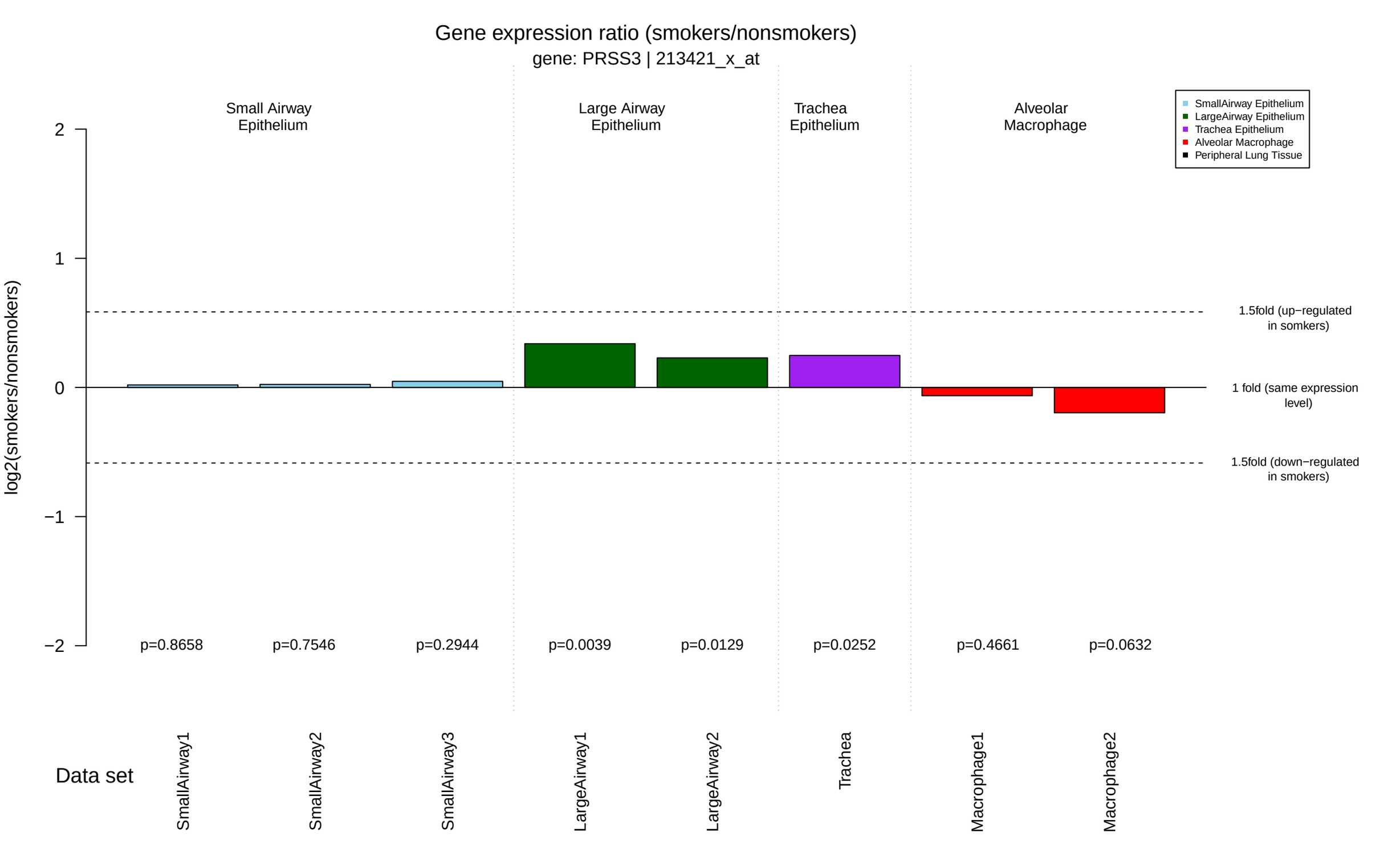
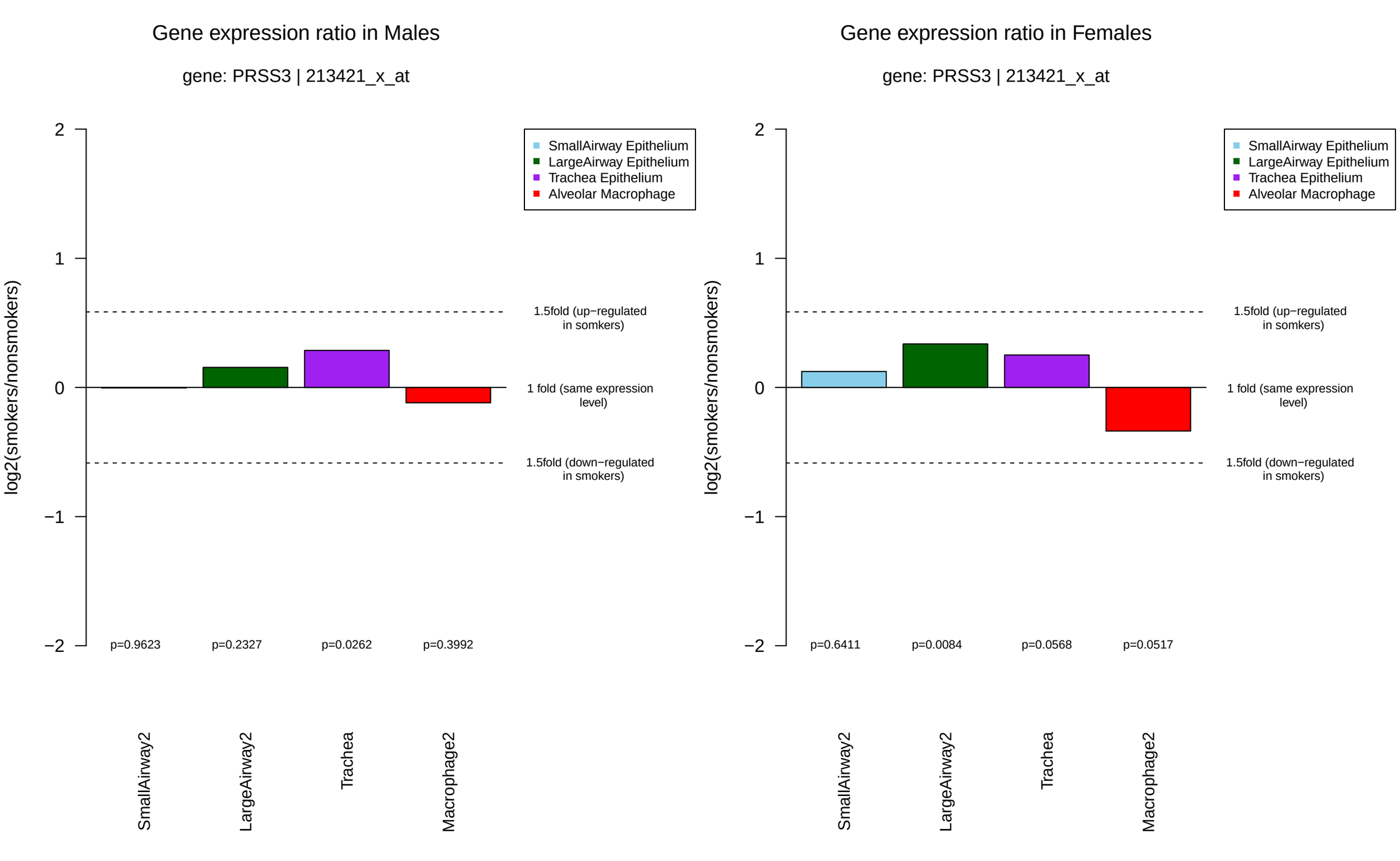
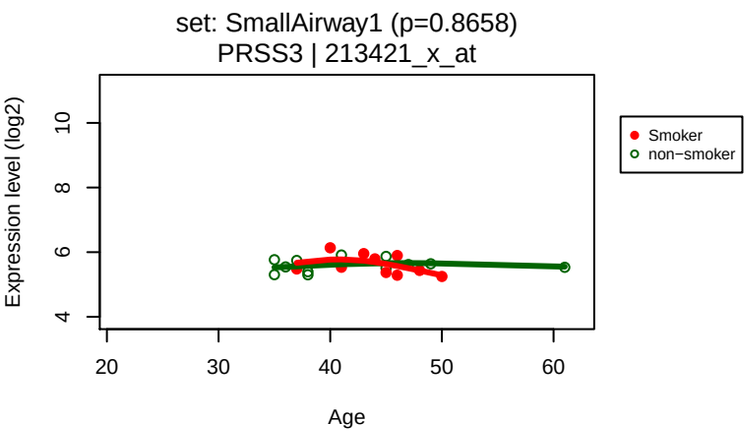
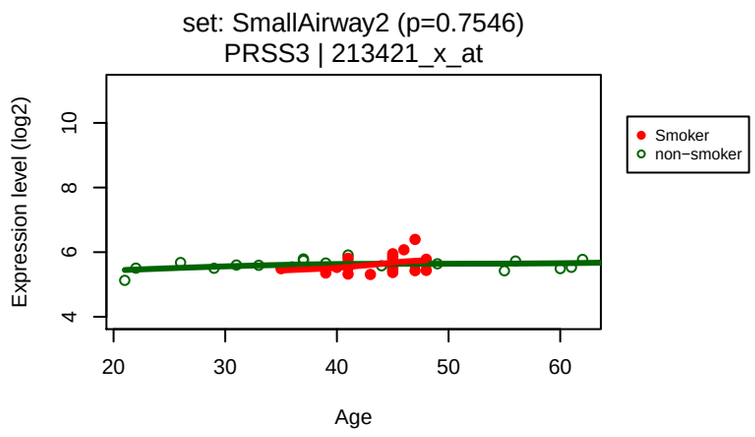
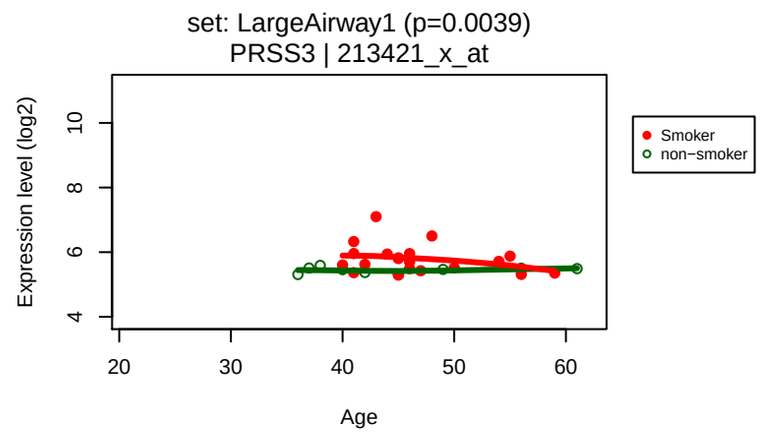

Gene expression of smokers and nonsmokers
gene: PRSS3 | 213421_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8658
p=0.7546
p=0.2944
p=0.0039
p=0.0129
p=0.0252
p=0.4661
p=0.0632
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

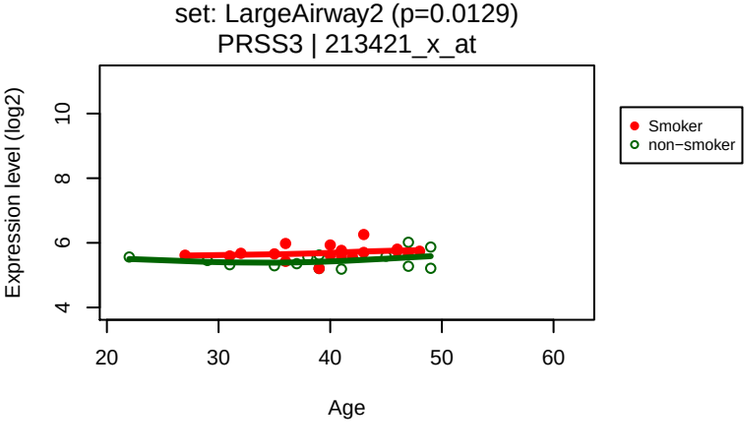

Gene expression ratio (smokers/nonsmokers)
gene: PRSS3 | 213421_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8658
p=0.7546
p=0.2944
p=0.0039
p=0.0129
p=0.0252
p=0.4661
p=0.0632
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

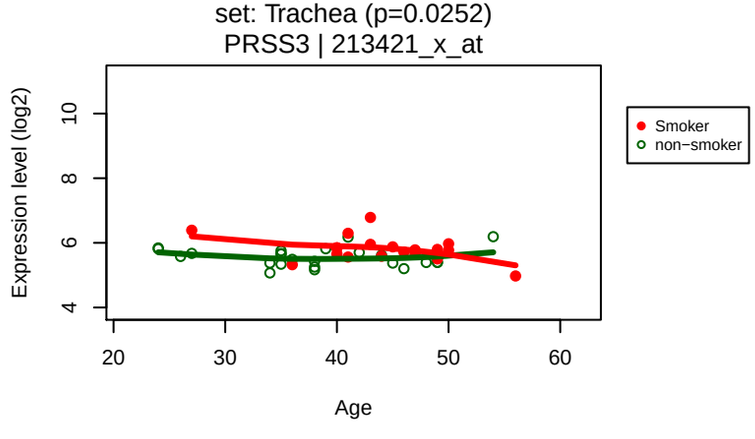

Gene expression ratio in Males
gene: PRSS3 | 213421_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9623
p=0.2327
p=0.0262
p=0.3992
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS3 | 213421_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6411
p=0.0084
p=0.0568
p=0.0517
SmallAirway2
LargeAirway2
Trachea
Macrophage2

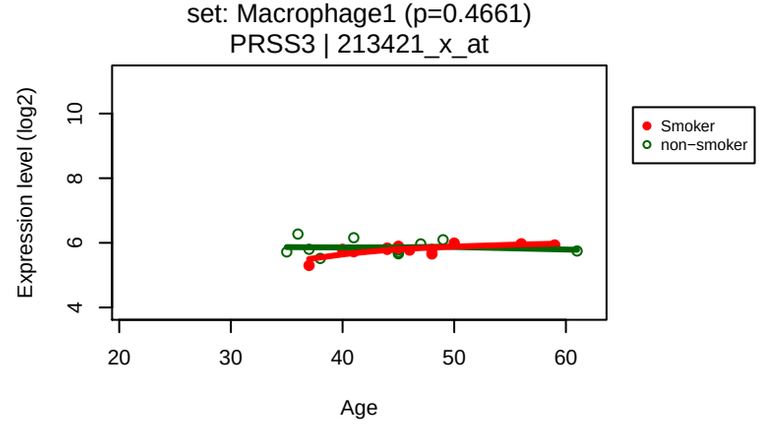

set: SmallAirway1 (p=0.8658)
PRSS3 | 213421_x_at
Expression level (log2)
Age
Smoker
non−smoker

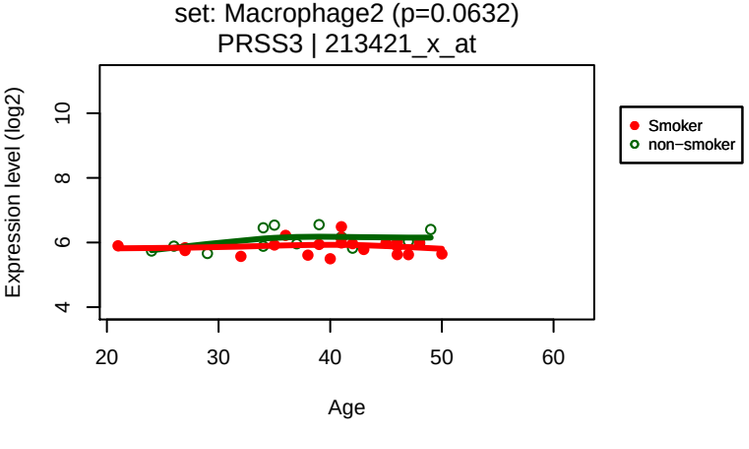

set: SmallAirway2 (p=0.7546)
PRSS3 | 213421_x_at
Expression level (log2)
Age
Smoker
non−smoker

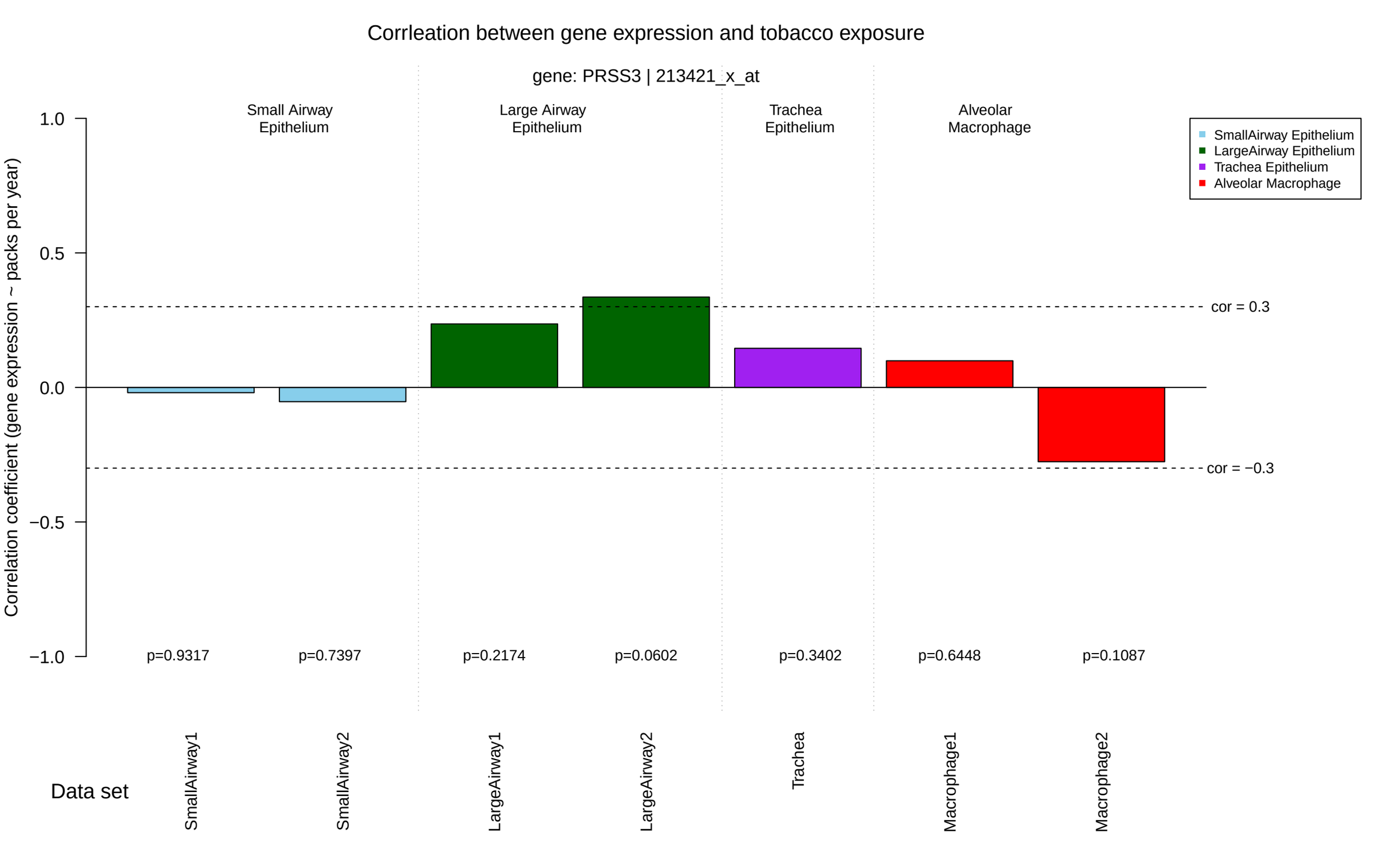

set: LargeAirway1 (p=0.0039)
PRSS3 | 213421_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0129)
PRSS3 | 213421_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0252)
PRSS3 | 213421_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4661)
PRSS3 | 213421_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0632)
PRSS3 | 213421_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS3 | 213421_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9317
p=0.7397
p=0.2174
p=0.0602
p=0.3402
p=0.6448
p=0.1087
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2